

network attachments, such as IP phones, wireless LAN access points and IP cameras, are designed as PDs.

Due to different power ranges needed by various PDs, IEEE802.3af defines an option to classify PDs according to their power consumption. Following the process of classification, a PD informs the PSE its power range so that the PSE can apply power more efficiently.

Class '0' is the default setting. A standard PSE which can recognise the class of the PD may supply power at the proper level to minimise the power usage and save the energy to another port. This not only cuts down power costs but also helps to eliminate the need of a cooling system.

Power-'on' procedure

To comply with both PoE-capable and non-PoE-capable devices, IEEE802.3af defines a method to prevent damage to an appliance because of improper power injection. According to the method, a PSE shall not apply power to a link until a PD is detected and requesting power. In addition, the PSE can supply power on different levels to meet the class which the PD belongs to. This power-'on' process involves signature detection and classification.

Signature detection. In this phase, the PSE tests whether a PD is connected. It applies two small current-limited DC volts across the cable and measures the impedance to determine whether a valid PD (with 25-kilo-ohm resistor) is present. When a PD is detected, power is applied. The PSE may further enter the optional classification phase to read the power level of the PD.

Classification. The PSE tries to classify a PD by presenting another voltage on the connection. The PD responds with a load current corresponding to its classification. The PSE can then identify the power requirement and optimise its power supply.

After the power-'on' process, the PSE raises the power level to fit the

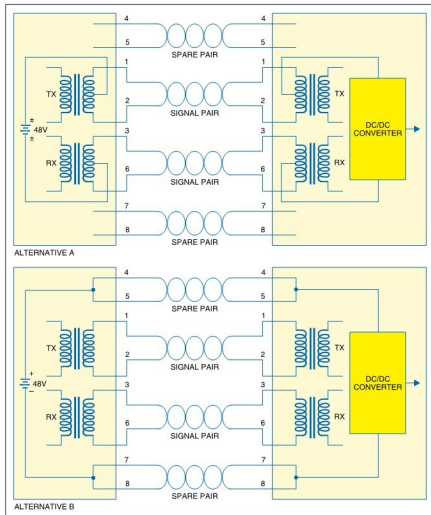


Fig. 2: Power and data arrangement on the cable

need of the PD and moves into 'on' state.

Disconnect protection

Once power is delivered, the PSE continuously monitors the current drawn by the PD for operational safety. If the draw drops below a minimum value or the user swaps a powered cable to a non-PoE device, the PSE stops supplying power to avoid damage. According to the standard, the PSE should cut the power once it loses the power signature from the PD in a very short time.

There are two detection schemes defined in the standard: DC disconnect and AC disconnect. The DC disconnect method removes power when the PD current falls below a

given threshold (5 to 10 mA) for a given time (300 ms to 400 ms). The AC disconnect superimposes a probing AC voltage on the power and measures the impedance. It removes power when the impedance is between 27 kilo-ohms and 1980 kilo-ohms and the AC voltage is found for a given time (300 ms to 400 ms).

PoE Plus

In 2004, the IEEE formulated a new group, IEEE802.3at, commonly known as Power over Ethernet Plus, to research the next generation of PoE. The main goal of the new standard is to increase the maximum power rating to 30 watts to fit the needs of higher-power-consuming applications such as WiMax base